

Name: _____



Securing understanding of arithmetic sequences

Task A: Reviewing the n^{th} term

1) Find the n^{th} term of these sequences.

a) $11, 19, 27, 35, 43, \dots$

b) $4, 12, 20, 28, 36, \dots$

c) $11, 7, 3, -1, -5, \dots$

d) $1.1, 1.9, 2.7, 3.5, 4.3, \dots$

e) $\frac{4}{11}, \frac{7}{11}, \frac{10}{11}, \frac{13}{11}, \frac{16}{11}, \dots$

Task B: Using the n^{th} term

1) For each sequence:

(i) find the n^{th} term,

(ii) find the 100th term,

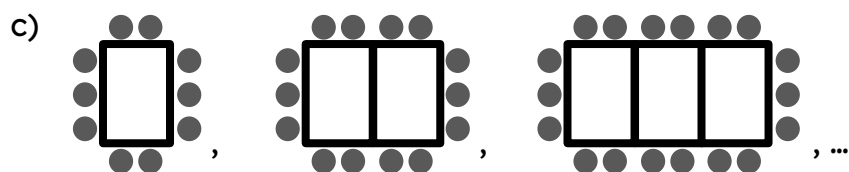
(iii) identify if 500 is in the sequence and if so, its position.

a) $-10, -7, -4, -1, 3, \dots$

b) $999, 996, 993, 990, 987, \dots$

c) $8.4, 8.8, 9.2, 9.6, \dots$

2) Assuming each pattern continues to grow the same way, will we see one with exactly 250 circles?



Task C: Generalising finding the n^{th} term

1) Write the first five terms of these sequences:

a) $5 + 4(n - 1)$

b) $4 + 5(n - 1)$

c) $4 - 5(n - 1)$

Hint: They are written in the form $a + d(n - 1)$ where a is the first term and d is the common difference.

2) Write two different generalisations for each of these arithmetic sequences:

a) $7, 10, 13, 16, 19, \dots$

b) $7, -3, -13, -23, -33, \dots$

c) $0.7, 0.95, 1.2, 1.45, 1.7, \dots$

d) $\frac{1}{7}, \frac{5}{7}, \frac{9}{7}, \frac{13}{7}, \frac{17}{7}, \dots$

3) For the sequence $24 + 7(n - 1)$

a) Find the first five terms.

b) Find the 100th term.

c) Find the position of the term 388 in the sequence.